Speedup Judge

`candidateIsFaster(reply, 0.02) → bool`

Input

Raw timings of candidate and Original SDK (-O2): 10 processes × 9 repetitions

Ratio

Per process: median candidate / median Original SDK; then median over processes

Test

Exact sign-flip test on the log-ratios

Return true

mean log-ratio < 0, p < 0.01, ratio ≤ 0.98

Return false = not confirmed faster.

Known speedup	Accepted / trials	95% CI of acceptance rate
1.000× (Null)	0/500	0–0.74%
≈1.010×	1/100	0.02–5.45%
≈1.020×	42/100	32.19–52.29%
≈1.050×	99/100	94.55–99.98%
≈1.100×	100/100	96.37–100%

Among tested points, detection reaches at least 90% from ≈1.05×.

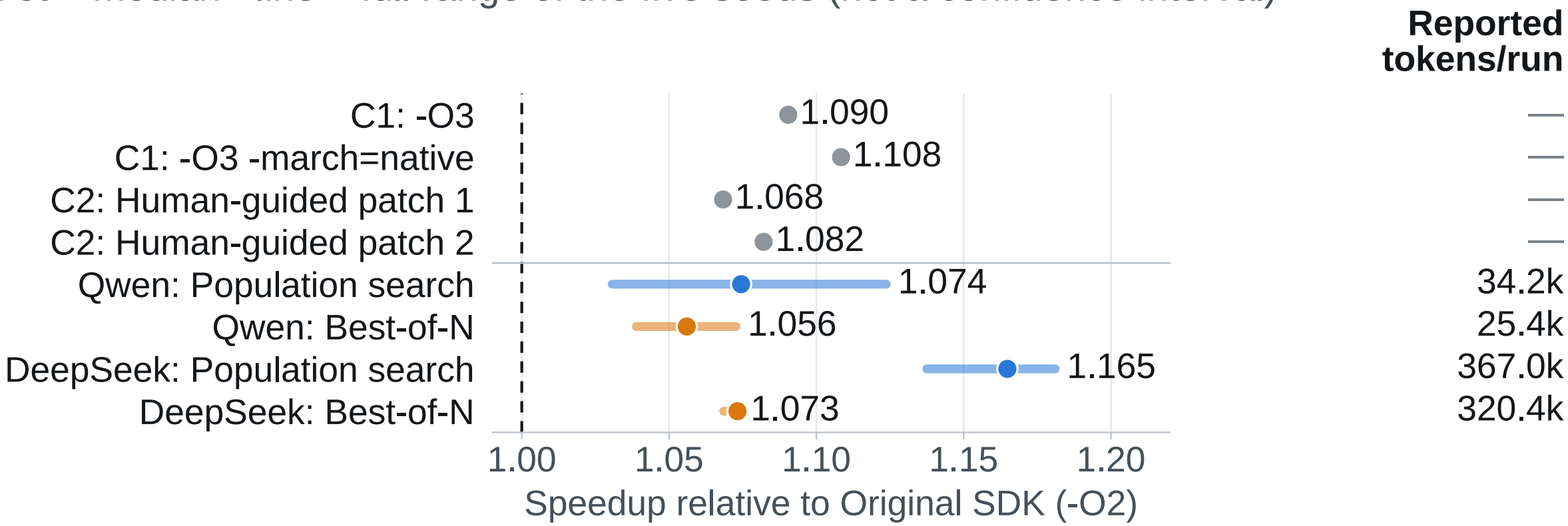
Largest absolute median-ratio deviation: 0.00145.

Search Results

Select a function in a parent patch → LLM rewrite → check and time → update population

C3: Best-of-N comparison · C4: 5 seeds, 40 requests per run

Dot = median · line = full range of the five seeds (not a confidence interval)



DeepSeek showed a clear observed search advantage; Qwen did not show a clear benefit in this sample.

Equal request budgets did not imply equal token costs.

Additional Correctness Tests

All 20 final patches passed the supplied correctness checks.

Additional test	Test coverage	Patches with new findings	Observed problem
Valid-input differential test	55 additional SDK-encoded files	**3/20**	Different decoded output
Mutated-input differential test	400 mutated files	**1/20**	More crashes than the Original SDK
ASan / UBSan	Up to 126 files, single thread	**1/20**	New memory or undefined-behavior reports
TSan	4 threads, up to 24 files	**15/20**	Data-race reports

A patch can have findings in more than one test.

The supplied checks did not cover all failures found by the broader tests.